

Optimal IPL Playing XI Selection and Batting Order Optimisation

Using Integer Programming and Goal Programming

Operations Research Project – First Review

Team A4

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Problem Context and Significance

IPL selection challenge

A team must choose only **11 players** from a larger squad while balancing:

- ▶ Batting strength.
- ▶ Bowling depth.
- ▶ Wicketkeeping.
- ▶ All-rounder availability.
- ▶ Batting-order suitability.

Objective

The objective is to:

- ▶ Select the **optimal playing XI**.
- ▶ Determine the **best batting order**.
- ▶ Balance batting and bowling strength.
- ▶ Ensure all cricket-related constraints are satisfied.
- ▶ Compare Integer Programming and Goal Programming solutions.

Historical IPL data → Player performance scores → Optimal XI + batting order

Problem Statement and Research Questions

Decision Problem

Select a feasible **Playing XI** from the candidate squad and assign each player a unique **batting position (1–11)**.

Key Decisions

- ▶ Select exactly **11 players**.
- ▶ Assign each player one batting position.
- ▶ Satisfy **team-composition constraints**.

Two Approaches

- ▶ **Integer Programming:** Maximise overall performance.
- ▶ **Goal Programming:** Minimise deviation from multiple targets.
- ▶ Both use the **same variables and constraints**.
- ▶ Identifies the **best overall scoring XI**

Dataset and Input Variables

Dataset

Source: Kaggle IPL dataset.

Coverage: Multiple IPL seasons, match-level and delivery-level records.

Raw variables

- ▶ Match ID, season, venue.
- ▶ Batting team and bowling team.
- ▶ Striker, bowler, over, ball.
- ▶ Runs, extras, wickets.
- ▶ Match winner and toss result.

Derived player inputs

- ▶ Batting average and strike rate.
- ▶ Boundary and dot-ball percentage.
- ▶ Economy rate and wickets.
- ▶ Recent-form score.
- ▶ Position-wise batting performance.
- ▶ Player role and bowling capability.

Common Decision Variables and Scores

Decision variables

$$x_i = \begin{cases} 1, & \text{if player } i \text{ is selected in the Playing XI} \\ 0, & \text{otherwise} \end{cases}$$

$$y_{ik} = \begin{cases} 1, & \text{if player } i \text{ bats at position } k \\ 0, & \text{otherwise} \end{cases}$$

Performance inputs

R_{ik} = expected batting contribution of player i at batting position k

$$S_i = w_B B_i + w_O O_i + w_F F_i + w_M M_i$$

B_i : batting score, O_i : bowling score, F_i : recent-form score, M_i : opponent/venue match-up score.

Approach 1: Integer Programming Model

Purpose

Select the XI and batting order that maximise the overall expected team contribution.

Objective function

$$\max Z_1 = \sum_{i \in P} \sum_{k=1}^{11} R_{ik} y_{ik} + \lambda \sum_{i \in P} O_i x_i$$

- ▶ The first term maximises expected batting contribution at appropriate positions.
- ▶ The second term rewards bowling capability in the selected XI.
- ▶ λ controls the importance assigned to bowling strength.

Output: Highest-scoring feasible Playing XI and batting order

Integer Programming Constraints

Select exactly 11 players

$$\sum_{i \in P} x_i = 11$$

Assign each selected player once

$$\sum_{k=1}^{11} y_{ik} = x_i \quad \forall i \in P$$

Fill each batting position

$$\sum_{i \in P} y_{ik} = 1 \quad \forall k = 1, \dots, 11$$

Role-balance constraints

$$\sum_{i \in WK} x_i \geq 1$$

$$\sum_{i \in BAT} x_i \geq 4$$

$$\sum_{i \in AR} x_i \geq 1$$

$$\sum_{i \in P} q_i x_i \geq 5$$

$q_i = 1$ if player i can bowl; otherwise $q_i = 0$.

Binary restrictions

$$x_i, y_{ik} \in \{0, 1\}$$

Approach 2: Goal Programming Model

Purpose

The same Playing XI and batting-order problem is solved by balancing several goals instead of maximising one composite score.

Goal equation

For each goal g :

$$f_g(x, y) + d_g^- - d_g^+ = T_g$$

- ▶ T_g : desired target value.
- ▶ d_g^- : under-achievement from the target.
- ▶ d_g^+ : over-achievement from the target.
- ▶ Only undesirable deviations are penalised.

Goal Programming objective

$$\min Z_2 = p_B d_B^- + p_O d_O^- + p_T d_T^- + p_F d_F^-$$

p_B, p_O, p_T, p_F represent priority/importance weights.

Goals Used in Goal Programming

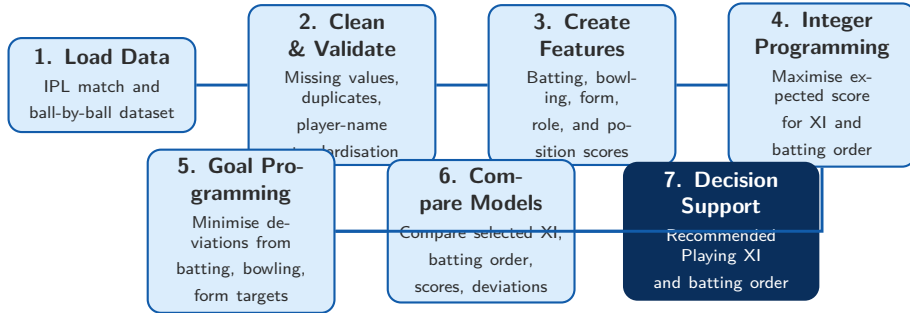
Goal	Target equation	Penalty
Batting strength	$\sum_{i,k} R_{ik} y_{ik} + d_B^- - d_B^+ = T_B$	d_B^-
Bowling strength	$\sum_i O_i x_i + d_O^- - d_O^+ = T_O$	d_O^-
Top-order quality	$\sum_i \sum_{k=1}^3 R_{ik} y_{ik} + d_T^- - d_T^+ = T_T$	d_T^-
Recent form	$\sum_i F_i x_i + d_F^- - d_F^+ = T_F$	d_F^-

Common hard constraints

The Goal Programming model retains all feasibility constraints from the Integer Programming model: exactly 11 players, one wicketkeeper, role balance, bowling coverage, and one player per batting position.

Proposed Methodology

Data Preparation → Player Scoring → Optimisation → Decision Support



Evaluation

Verify role constraints and batting-position assignments; compare total score, goal deviations, role balance, and sensitivity to changed weights or targets.

Proposed User Interface

IPL Team Optimisation Dashboard

- 1 Load the IPL dataset.
- 2 Choose a squad, opponent, venue, or season.
- 3 View player statistics and role distribution.
- 4 Set constraints and performance targets.
- 5 Run **Integer Programming**.
- 6 Run **Goal Programming**.
- 7 Compare selected XI, batting order, and score breakdown.

IPL TEAM OPTIMISATION DASHBOARD

Load Data View Statistics Select Squad

Run Integer Programming Run Goal Programming

[Selected XI](#) | [Batting Order](#) | [Constraint Status](#) | [Model Comparison](#)

Tools, Evaluation, and Expected Outputs

Tools

- ▶ Python, Pandas, NumPy.
- ▶ PuLP / Pyomo / OR-Tools.
- ▶ Streamlit user interface.
- ▶ Matplotlib or Plotly.
- ▶ Jupyter Notebook and GitHub.

Evaluation

- ▶ Constraint satisfaction.
- ▶ Batting and bowling scores.
- ▶ Role balance.
- ▶ Sensitivity to weights/targets.
- ▶ Difference between IP and GP line-ups.

Expected outputs

- ▶ Analysis-ready IPL dataset and statistics.
- ▶ Two optimisation implementations for the same decision problem.
- ▶ Selected XI and batting-order recommendations.
- ▶ Interactive comparison dashboard.

Conclusion and Next Steps

Project contribution

- ▶ Formulates IPL Playing XI selection and batting-order assignment as one Operations Research problem.
- ▶ Applies **Integer Programming** and **Goal Programming** to the same decision.
- ▶ Compares maximum-performance and target-balanced team-selection strategies.
- ▶ Delivers a data-driven decision-support interface.

Next steps

Dataset finalisation → statistical analysis → score construction → model implementation → UI development → sensitivity analysis

First-review takeaway: A clear OR formulation for selecting an IPL XI and batting order.

Thank You

Questions?

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